

Ahmad Al-Imtiaz

Graduate Research Assistant, CASSA, IUB

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EDUCATION

Master of Science in Physics [CGPA: 3.56 (out of 4)] **Nov, 2024**

Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh

Bachelor of Science in Physics [CGPA: 3.28 (out of 4)] **Oct, 2022**

Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh

RESEARCH INTEREST

Extragalactic astronomy, galaxy formation and evolution, chemical abundances and spectroscopy, early universe, observational cosmology, studies of galaxies and cosmology via gravitational lensing, computational astrophysics, and observational astrophysics.

RESEARCH EXPERIENCE

Publications

- Adnan, S. M. R.†, Hasan, M. J. †, **Imtiaz, A. A.†**, Robin, S. H.†, Shwadhin, F. R.†, Shajib, A. J., Nahid, M. H., Tanver, M. H., Akter, T., Jahan, N., Jafar, Z., Rashid, M., Biswas, A., Chowdhury, A. A., Feardous, J., Rahaman, A., Ridwan, M., Sharma, R. D., Chowdhury, Z., Hossain, M. S.. [Investigating the relation between the environment and internal structure of massive elliptical galaxies using strong lensing | Astronomy & Astrophysics](#) [† these authors contributed equally to this work.]

Active Projects

Metallicity gradients estimation of the stellar populations on AGEL0014 **Preparing Manuscript**

I am analyzing the stellar population of the lensed quiescent galaxy AGEL0014 using data from Gemini/GSAOI, HST, and Keck/MOSFIRE. The work involves image reduction using *dragons*, lens modeling with *Lenstronomy*, spectral reduction (*PyPeIt*), and spectral fitting (*alf*). Intensive computations are being performed on the HPC cluster at NARIT. [I will be the first author in this work.]

Supervisor: **Dr. Nicha Leethochawalit, NARIT, Thailand.**

Detailed study of star formation history of the Sparkler galaxy body **Expected: Jun 26**

I am using JWST/NIRSpec IFU spectroscopic data and lens modeling (*Lenstronomy*) to study the Sparkler galaxy ($z=1.4$) and its ‘sparkles,’ candidate ancient globular clusters. By combining multi-wavelength data from JWST and HST with SED fitting (*Pixedfit*, *DenseBasis*), I aim to constrain their ages, metallicities, and the host galaxy’s star formation history.

Supervisor: **Dr. Lamiya Mowla, Wellesley College, USA.**

SKILLS

Research Tools: Mathematica, GitHub, Overleaf, ADQL, TOPCAT, APT, ds9, dragons (GSAOI-reduction pipeline), Lenstronomy, Dense Basis, alf, MS Office 365

Programming language: Python, C, MATLAB

Operating Systems: Windows, Linux

Languages: Bengali (Native), English (IELTS-6.5)

Telescopes:

- Celestron CGEM II 800 SCT - 8" on CGEM II Computerized Equatorial Mount
- EvoStar 80ED Apo Refractor on MEADE LX85 Computerized Equatorial Mount

PROFESSIONAL & OTHER EXPERIENCES

- **Project Guide and Guarantor**, EcoAstroBD, funded by **IAU-OAD** Dec 2025 - Present
A youth-led initiative in Bangladesh to reduce educational inequalities in STEM by developing and distributing affordable, eco-friendly astronomy kits.
- **Graduate Research Assistant**, **CASSA**, IUB (*Full-time*) [Contractual] Nov 2025 – Present
Working on “star formation history of the Sparkler galaxy body” under Dr. Mowla’s remote supervision.
- **Research Intern**, ICTP PWF Bangladesh: Summer Internship Program [Remote] Jul - Oct 2025
- **SOC & Instructor**, **CAM-SUST Summer School on Astrophysics 2025** Aug-Sep 2025
- **Instructor, Stellar Light & Cosmic Bends: Data-Driven Astronomy Workshop** 27-28 July 2025
Led a hands-on session on galaxy–galaxy strong lens modeling.
- **Lecturer, Dept. of CSE, Daffodil International University** (*Full-time*) [Contractual] Feb-Apr 2025
I taught PHY101 (Mechanics, Waves and Oscillations, Thermodynamics, Optics), PHY102 (Electricity and Magnetism, Modern Physics), and PHY103 (Lab), which are junior undergraduate-level courses.
- **Asst. Teacher – Science (Middle School)** (*Full-time*) [Permanent] Apr 2024–Feb 2025
Member of the Corporate Social Responsibility team at Epyllion Group, Dhaka
- **Instructor**, AstroCode Intensive, **CAM-SUST** Mar 08–09, 2024
Conducted a hands-on session on astronomical image reduction techniques.
- **President**, 9th Executive Committee, **CAM-SUST** Mar 2021–Jul 2022
- **Study Circle Secretary**, 8th Executive Committee, CAM-SUST Aug 2019–Mar 2021

CONFERENCE

- “Exploring the Mass-Metallicity Relation in Open Clusters inside the Milky Way” Nov 13–17, 2023
Poster at Metal Production and Distribution in a Hierarchical Universe – II, ESO Santiago, Chile
- “Stellar Multiplicity-dependent Neptunian Desert Boundary” Sep 23–26, 2022
Poster at International Conference of Physical Sciences by School of Physical Sciences, SUST

LEARNING EXPERIENCES

Schools and Workshops

- **CASSA Workshop 1: Unclouding the Thousand Eyes of an Array Radio Telescope, 2025 [In-person]**
- **JWST Data Analysis and Processing Workshop (South East Asia) 2024 [In-person]**
Organized by COSPAR & NARIT, CHIANG MAI, THAILAND
- **Astro101: Hands on Workshop on Optical Astronomy 2023 (Hackathons Winner) [In-person]**
Organized by NARIT, CHIANG MAI, THAILAND
- **Introduction to Computational Astrophysics, 2021** (NUEVACAMP, SPSB, MASLAB, BANGLADESH)
- **Introduction to HR Diagram with LCO (2021)**, Prof. S N Hasan and Dr Priya Hasan
MAULANA AZAD NATIONAL URDU UNIVERSITY, INDIA
- **HPC for Astrophysics and Astronomy, 2021** (SKA- INDIA CONSORTIUM & IIT, KHARAGPUR)
- **Introduction to MATLAB, 2021** (IEEE NSU PES, STUDENT BRANCH CHAPTER)
- **Introduction to Stars, 2021** (BREAKTHROUGH SCIENCE SOCIETY INDIA)
- **Machine Learning, 2021** (Andrew Ng, Coursera)
- **13th Radio Astronomy Winter School 2020** (IUCAA, PUNE, INDIA)
- **Astronomical Data Analysis by TOPCAT, 2019** (CAM-SUST, Sylhet, Bangladesh)

Competitions

- **15th position** in [International Theoretical Physics Olympiad for Undergraduate Students, 2019](#) (Online)
- **Winner** in University Physics Olympiad 2019, organized by CAM-SUST, Sylhet, Bangladesh (In-person)
- Accomplished Competitor in [The University Physics Competition 2018](#) (Online)
- Accomplished Competitor in [The University Physics Competition 2019](#) (Online)

OUTREACH

Volunteer & consultant , CAM-SUST, Astro-Carnival, SUST (More than 700 Students attended)	Sep 2023
Volunteer , CAM-SUST, Astronomy School Program, Jaflong (Around 200 students attended)	Mar 2022
Convener , CAM-SUST, Astronomy School Program, Sunamganj (Around 130 Students attended)	Mar 2020
Volunteer , CAM-SUST, Astronomy School Program, Sonargaon (Around 150 students attended)	Jan 2020
Convener , CAM-SUST, Astronomy School Program, Brahmanbaria (About 500 Students attended)	Oct 2019
Volunteer , Physics Olympiad, National Round, BDPhO (Around 700 Students attended)	Mar 2019

Oral Presentation

- Image Reduction on an off-line Study Circle of CAM-SUST Jan 2024
- Introduction to Stars- II on an off-line Study Circle of CAM-SUST Mar 2023
- Introduction to Stars- I on an online Study Circle of CAM-SUST Oct 2022
- Celestial Coordinate System on an online Study Circle of CAM-SUST Oct 2020
- Detection of Exoplanets, at a School Program Organized by CAM-SUST Oct 2019

REFERENCES

Dr. Anowar J. Shajib

NFHP Einstein and KICP Fellow,
University of Chicago
Email: ajshajib@uchicago.edu

Dr. Nicha Leethochawalit

Researcher,
National Astronomical Research Institute of Thailand
Email: nicha@narit.or.th

Dr. Lamiya Mowla

Assistant Professor of Astronomy,
Wellesley College, USA
Email: lmowla@wellesley.edu